

A Boundary Value Problem Approach to Autonomous Rendezvous in Industrial Human-Drone Interaction

Silvia Proia* Raffaele Carli** Graziana Cavone***
Mariagrazia Dotoli** Lorenzo Sabattini*

* *Department of Sciences and Methods for Engineering, University of Modena and Reggio Emilia, Italy (e-mail: {silvia.proia, lorenzo.sabattini}@unimore.it).*

** *Department of Electrical and Information Engineering, Polytechnic of Bari, (e-mail: {raffaele.carli, mariagrazia.dotoli}@poliba.it)*

*** *Department of Civil, Computer Science, and Aeronautical Technologies Engineering, University Roma Tre, Italy (e-mail: graziana.cavone@uniroma3.it)*

Abstract: This paper addresses the challenge of human–drone rendezvous in industrial environments by presenting a novel boundary value problem formulation that, unlike existing literature, incorporates a receding horizon strategy to adapt to human behavioral variability. The approach addresses in an integrated way safety, ergonomics, and efficiency objectives by an optimal control strategy based on the Pontryagin Maximum Principle. A shrinking-horizon scheme enables real-time adaptation to human motion, while analytically satisfying terminal boundary conditions. Safety is ensured through a distance-dependent speed law, and ergonomic feasibility is imposed as a spatial constraint on the rendezvous point consistent with the Rapid Upper Limb Assessment method. Energy and motion efficiency are achieved by penalizing control effort while enforcing trajectory smoothness and dynamic feasibility. Simulations in a realistic warehouse scenario show that the proposed strategy enables safe and ergonomic cooperation with a moving operator, achieving improved control performance and terminal convergence compared with an existing receding-horizon approach.

Keywords: Automatic control, optimization, real-time operations in transportation, autonomous mobile robots, human-drone interaction, ergonomics, safety, efficiency, Pontryagin Maximum Principle, receding horizon boundary value problem

1. INTRODUCTION

The progressive digitalization of industrial environments and the advent of Industry 4.0 are fostering the integration of autonomous aerial systems within human-centered production processes. Uncrewed Aerial Vehicles (UAVs) are increasingly deployed to support pick-and-delivery, inspection, and transportation operations, enabling greater flexibility and operational efficiency in collaborative logistics (Wawrla et al. (2019)). The coexistence of aerial robots and human operators in tight indoor environments, however, introduces new challenges related to motion coordination, workspace sharing, and the preservation of human well-being. Within this context, Human–Drone Interaction (HDI) represents a research domain that extends the principles of Human–Robot Interaction (HRI) to aerial

systems, aiming to guarantee safe, ergonomic, and efficient cooperation during industrial tasks (Proia et al. (2025)).

Recent literature highlights that HDI has been predominantly investigated in outdoor or service-oriented domains, including photography, surveillance, and inspection (Mirri et al. (2019)), whereas contributions focused on indoor industrial environments remain limited (Proia et al. (2025)). The extension of HDI to industrial contexts is motivated by the objective of enhancing automation and operational flexibility while preserving operator safety, comfort, and energy efficiency. Safety is commonly addressed through operational methodologies derived from collaborative robotics, such as Speed and Separation Monitoring and Power and Force Limiting (Proia et al. (2021); Lucci et al. (2020)), which define motion and proximity constraints in shared workspaces. Ergonomics aspects play an equally relevant role, aimed at reducing biomechanical strain and maintaining postural comfort as key factors for sustainable human–robot coexistence. Quantitative assessment procedures, including the Rapid Upper Limb Assessment (RULA) (McAtamney and Corlett (1993)), are extensively employed to evaluate operator’s physical loads and to delineate ergonomically admissible interac-

* The work of R. Carli and M. Dotoli was supported in part by the National Recovery and Resilience Plan (NRRP), Mission 4 Component 2 Investment 1.3 - Call for tender No. 341 of March 15 2022 of Italian Ministry of University and Research (funded by the European Union – NextGenerationEU) under the project “MICS (Circular and Sustainable Made-in-Italy)” (project code: PE000000004).

tion regions (Proia et al. (2023, 2025)). Efficiency is an additional requirement in HDI, as it directly influences productivity and energetic sustainability of autonomous operations. Designing energy-efficient trajectories allows the drone to complete tasks with minimal power consumption, enhancing overall performance in human-centered industrial environments.

In pick-and-delivery operations, the collaborative descent of a drone toward a moving operator represents a critical phase of HDI. The aerial platform must maintain safe proximity and precise spatial alignment with the operator, while ensuring dynamically feasible motion, ergonomics compliance, and energetic efficiency at the delivery configuration. The corresponding planning and control problem can be formalized as an *autonomous rendezvous*, in which the drone trajectory is required to converge to a terminal state consistent with the human motion.

In the literature, rendezvous and landing problems have generally been expressed as optimal control or trajectory-tracking formulations. Classical Proportional-Integral-Derivative (PID) and Linear Quadratic Regulator (LQR) controllers provide satisfactory performance under linearized dynamics but do not guarantee the explicit enforcement of terminal boundary conditions and exhibit sensitivity to parameter tuning in nonlinear regimes (Wei et al. (2025); Proia et al. (2025)). Model Predictive Control (MPC) has become one of the most widely adopted strategies for rendezvous applications, due to its ability to handle multi-constraint optimization in real-time. Indeed, several contributions have demonstrated the effectiveness of MPC in autonomous docking and target-tracking problems, including spacecraft and aerial rendezvous scenarios (Li et al. (2017); Quartullo et al. (2024)). Despite its flexibility, the finite prediction horizon and the penalization of terminal states in MPC formulations prevent the exact satisfaction of terminal constraints, limiting convergence accuracy in rendezvous maneuvers. Direct transcription techniques yield accurate discretized representations of the optimal control problem but lead to accumulating approximation errors due to mesh discretization, often producing discontinuities in the control profile and suboptimal terminal accuracy (Luo et al. (2014); Proia et al. (2023)).

A more rigorous formulation of the rendezvous problem is obtained through the indirect solution of the optimal control problem derived from the *Pontryagin Maximum Principle* (PMP) (Kopp (1962)). In particular, coupling state and costate dynamics through the Hamiltonian minimization condition transforms the rendezvous into a *two-point Boundary Value Problem* (BVP) (Miele et al. (2007); Avvakumov and Kiselev (2000)), where both initial and terminal constraints are satisfied exactly. In this framework, both safety and efficiency are embedded in the cost function, the former through penalization terms in the Hamiltonian structure and the latter through drone’s control energy minimization, whereas ergonomic feasibility is imposed via terminal boundary conditions defining admissible human-drone configurations. The resulting system of differential equations is solved using a collocation method (Wang (2009)), which ensures fourth-order accuracy and continuity of the control profile. From a methodological perspective, the present work contributes the literature with an extension of the BVP to a *receding-horizon* for-

mulation, introducing a real-time adaptive control strategy for *human-drone rendezvous*. To the best of the authors’ knowledge, no previous study has applied a *shrinking-horizon* BVP scheme to optimal rendezvous. The main advantages of the proposed formulation is that it preserves the analytical rigor of indirect optimal control while embedding a feedback mechanism that iteratively reformulates and solves the BVP over a progressively shorter horizon, allowing continuous adaptation of the trajectory to changes in the task configuration and workspace geometry. In contrast to conventional MPC and direct transcription methods, the receding-horizon BVP guarantees exact terminal convergence through analytical enforcement of boundary conditions while retaining the adaptability typical of predictive schemes. Treating the final time as a free variable generates efficient trajectories consistent with the relative dynamics of the drone and operator, while the global coupling of state and costate variables enables integrated optimization of the entire motion rather than local stepwise corrections.

Building on the receding-horizon BVP formulation, the proposed HDI control framework provides a unified approach for indoor industrial scenarios, combining the precision of indirect optimal control with the efficiency required for real-time operation. Compared with the descent control approach proposed in Proia et al. (2025), the present formulation enforces terminal constraints exactly, achieving higher convergence accuracy and enabling energy-efficient, safety- and ergonomically-compliant interaction dynamics.

Summarizing, the main contributions of this paper are:

- Formulation of an autonomous rendezvous control strategy for HDI in indoor industrial environments, integrating safety, ergonomics, and efficiency considerations into a unified optimal control framework;
- Development of a shrinking-horizon BVP algorithm, allowing real-time adaptation to evolving interaction conditions while preserving exact satisfaction of boundary constraints.

The remainder of this paper is structured as follows. Section 2 presents the system modeling and HDI operating framework. Section 3 formulates the rendezvous optimal control problem, whereas Section 4 details its reformulation into the indirect BVP representation. Section 5 introduces the proposed shrinking-horizon BVP-based control strategy for adaptive human-drone rendezvous. The numerical setup and results are discussed in Section 6. Finally, concluding remarks are reported in Section 7.

2. SYSTEM MODELING AND HDI OPERATING FRAMEWORK

This section details the drone dynamics (Section 2.1), the human representation (Section 2.2), and the industrial HDI scenario and problem statement (Section 2.3). Among different UAVs, the drone considered in this work is a quadrotor, representative of the class of multirotor aerial vehicles commonly employed in indoor environments due to their maneuverability and compact design.

2.1 Quadrotor Dynamics

A quadrotor, also referred to as a quadcopter, consists of four rotors arranged in two counter-rotating pairs

at the corners of a rigid square frame. In the context of HDI, the inner attitude control loop ensures near-hover stability, enabling a clear separation between the slow translational dynamics and the fast rotational one. Accordingly, the following formulation focuses on the translational dynamics of the drone, modeled as a three-degree-of-freedom system evolving along the Cartesian axes of the East-North-Up (ENU) inertial reference frame, while the orientation control is assumed to maintain nominal hover conditions.

The quadrotor is modeled as a symmetric rigid body under small-angle assumptions, neglecting aerodynamic drag, propeller-airframe coupling, and ground effects, as outlined in Sabatino (2015) and Das et al. (2009).

The state vector of the drone is defined as

$$s^d(t) = (p^d(t)^\top \dot{p}^d(t)^\top)^\top \in \mathbb{R}^6, \quad (1)$$

where $p^d(t) = (x^d(t), y^d(t), z^d(t))^\top$ denotes the position in the inertial frame, and $\dot{p}^d(t) = (\dot{x}^d(t), \dot{y}^d(t), \dot{z}^d(t))^\top$ its linear velocity, compactly indicated as $v^d(t) := \dot{p}^d(t)$.

The control input is expressed as

$$u^d(t) = (a_x^d(t) \ a_y^d(t) \ a_z^d(t))^\top \in \mathbb{R}^3, \quad (2)$$

representing the equivalent translational accelerations along the body axes, generated by the attitude control layer to regulate the drone motion in the inertial frame.

Employing the state vector $s^d(t)$ and the control vector $u^d(t)$, the translational dynamics is expressed as

$$\begin{aligned} \dot{s}^d(t) &= f^d(s^d(t), u^d(t), t) \\ &= \begin{bmatrix} \dot{p}^d(t) \\ -\frac{k^d}{m^d} \dot{p}^d(t) + \frac{1}{m^d} u^d(t) \end{bmatrix} \\ s^d(t_0) &= s^{d,0}. \end{aligned} \quad (3)$$

where m^d is the drone mass and $k^d > 0$ is a viscous damping coefficient accounting for aerodynamic resistance. The function $f^d : \mathbb{R}^{10} \rightarrow \mathbb{R}^6$ represents the nonlinear, time-varying translational dynamics of the quadrotor, and $s^{d,0} \in \mathbb{R}^6$ denotes the initial state of the drone at the initial time t_0 .

The explicit time dependence in (3) allows the model to capture nonstationary operating conditions, such as adaptive speed modulation or time-varying environmental effects arising in HDI scenarios.

2.2 Human Representation

The human operator moves along a smooth, time-parametrized trajectory that serves as an external input to the rendezvous problem. We assume that the operator's position evolves as $p^h(t) = (x^h(t), y^h(t))^\top \in \mathbb{R}^2$, whose time derivative $v^h(t) := \dot{p}^h(t) = (\dot{x}^h(t), \dot{y}^h(t))^\top$ represents the instantaneous velocity. The operator's heading (i.e., the orientation of the body with respect to the global frame) $\psi^h(t)$ is assumed to remain aligned with the instantaneous direction of motion, meaning that the operator faces and moves forward along the trajectory without backward or lateral motion (i.e., $\psi^h(t) =$

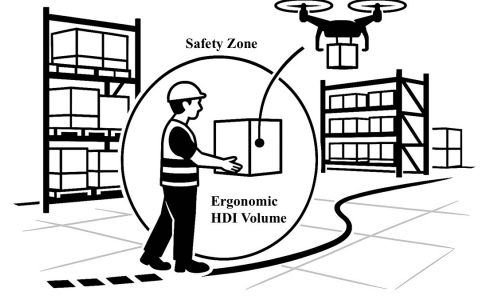


Fig. 1. Safe, ergonomic, and efficient rendezvous in industrial HDI.

$\text{atan2}(\dot{y}^h(t), \dot{x}^h(t))$). The explicit modeling of human biomechanics, motion prediction, and behavioral variability is considered beyond the scope of our work.

2.3 HDI Scenario and Problem Statement

We consider an industrial warehouse environment where drones operate in close collaboration with human workers for pick-and-delivery tasks. A quadrotor retrieves a package from a storage shelf and, starting from a hovering position, descends to transport the payload toward a designated operator moving within the palletizing area (see Fig. 1).

The operational context is characterized by industrial HDI, emphasizing the integration of aerial robots into collaborative logistics processes. The interaction takes place within a well-delimited industrial area, where human presence, workspace layout, and mandatory safety regulations impose strict operational constraints. The drone motion must therefore account for the operator's movement and the geometric limitations of the shared workspace. During the drone's descent phase, dynamic safety conditions regulate approach speed, ensuring collision-free operation and compliance with cooperative workspace boundaries. The final delivery configuration must lie within an ergonomically admissible region identified through a postural comfort analysis based on the RULA index (Proia et al. (2025)), ensuring that the resulting motion remains both energy-efficient and ergonomically safe.

In this context, HDI is treated as a control problem aimed at determining a dynamically feasible drone trajectory and terminal configuration that enable an *autonomous rendezvous under safety, ergonomics, and efficiency compliance*. The objective is to ensure safe proximity to the operator, preserve postural comfort, and achieve energy-efficient and sustainable HDI within the industrial environment.

3. RENDEZVOUS PROBLEM FORMULATION

The rendezvous trajectory planning problem for HDI is formulated in continuous time as a free-final-time optimal control problem. The objective is to minimize actuation energy and translational kinetic effort while ensuring proximity safety and ergonomic feasibility of the final interaction configuration.

Let $[t_0, t_f]$ denote the finite time horizon, where $t_0 \in \mathbb{R}$ is the initial time and $t_f \in \mathbb{R}$ is the terminal time. The

horizon is finite, but the terminal time t_f is a free variable to be determined as part of the optimization process. The continuous time variable is denoted by $t \in [t_0, t_f]$.

The overall formulation, presented in the following, integrates three essential components: (i) a safety regulation law ensuring distance-based speed modulation (Section 3.1); (ii) an ergonomic constraint defining the admissible terminal configuration (Section 3.2); and (iii) energy and motion efficiency terms promoting smooth and energy-aware trajectories (Section 3.3).

All components are integrated within the optimal control problem presented in Section 3.4, combining safety regulation, ergonomic feasibility, and energy efficiency.

3.1 Safety Regulation

To ensure safe HDI, the drone motion must comply with a minimum protective distance from the operator and adapt its speed to the instantaneous separation distance. We define a continuous safety regulation law that prescribes a time-varying *safe speed profile* $v_{\text{safe}}(t)$, modeled as a nonincreasing function of the relative positions of the two agents:

$$v_{\text{safe}}(t) = \tilde{v}_{\text{safe}}(p^d(t), p^h(t)) = \begin{cases} 0, & d(t) \leq d_{\text{stop}}, \\ v_{\text{cruise}} \frac{d(t) - d_{\text{stop}}}{d_{\text{cruise}} - d_{\text{stop}}}, & d_{\text{stop}} < d(t) < d_{\text{cruise}}, \\ v_{\text{cruise}}, & d(t) \geq d_{\text{cruise}}, \end{cases} \quad (4)$$

where the instantaneous human-drone separation is defined as

$$d(t) = \|p^d(t) - p^h(t)\|, \quad (5)$$

and the parameters d_{stop} and d_{cruise} denote respectively the inner and outer bounds of the protective region, while $v_{\text{cruise}} > 0$ represents the nominal cruise speed. The profile ensures smooth and monotonic deceleration as the drone approaches the operator, avoiding abrupt speed changes and maintaining dynamic feasibility. Safety regulation is integrated into the cost function through a quadratic penalty term based on $v_{\text{safe}}(t)$, enabling the optimization process to continuously adapt the drone speed while ensuring compliance with proximity requirements.

3.2 Ergonomic Feasibility

Ergonomic feasibility defines the spatial configurations that allow the drone to approach the human operator under admissible postural effort. The formulation extends the ergonomic model proposed by Proia et al. (2025), which employs the RULA index to quantify the postural condition as a function of the human-drone relative pose. An ergonomic volume $\mathcal{V}^* \subseteq \mathbb{R}^3$ is defined within the collaborative work zone, representing the set of spatial configurations corresponding to the minimum RULA index that can be comfortably reached by the operator.

The volume is expressed in the human reference frame $\{F^h(t)\}$, rigidly attached to the body with origin at the position $p^h(t)$, located between the operator's feet. The axes are aligned with the anthropometric directions, with

x oriented forward along the torso, y directed toward the left arm, and z oriented vertically upward. In the $\{F^h(t)\}$ coordinates, the ergonomic volume $\mathcal{V}^*$ lies in front of the torso and is slightly shifted toward the operator's dominant-hand side, corresponding to the natural reaching area for frontal manipulation.

The relative configuration between the drone and the operator within $\{F^h(t)\}$ is described by the ergonomic displacement vector

$$\delta = (\delta_x, \delta_y, \delta_z)^\top \in \mathcal{V}^* \subset \mathbb{R}^3, \quad (6)$$

where δ_x is the forward offset relative to the torso, δ_y is the lateral offset toward the operator's dominant-hand side, and δ_z is the vertical offset relative to the human frame origin.

Since the rendezvous control problem is formulated in the ENU inertial reference frame, a coordinate transformation maps the ergonomically admissible region into the same frame:

$$p^d(t_f) = p^h(t_f) + R^h(t_f) \delta, \quad \delta \in \mathcal{V}^*, \quad (7)$$

where $R^h(t_f) \in SO(3)$ denotes the rotation matrix describing the orientation of $\{F^h(t)\}$ with respect to the ENU frame, given by

$$R^h(t_f) = \begin{bmatrix} \cos \psi^h(t_f) & -\sin \psi^h(t_f) & 0 \\ \sin \psi^h(t_f) & \cos \psi^h(t_f) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (8)$$

The combined effect of translation and rotation preserves the geometry of the ergonomic volume in the human frame and ensures a consistent positioning in the ENU frame as the operator moves.

Accordingly, the terminal configuration ensuring ergonomic admissibility is defined by (7), which constrains the drone's final position to lie within the ergonomically admissible region $\mathcal{V}^*$. A rest-to-hover condition is further imposed as

$$\dot{v}^d(t_f) = 0. \quad (9)$$

3.3 Energy and Motion Efficiency

Energy consumption represents a key aspect of HDI, as it directly influences operational efficiency and mechanical stress on the aerial platform. In the proposed framework, energy efficiency is achieved by penalizing the control effort, while additional kinematic terms promote smooth and dynamically feasible motion, as detailed in Section 3.4.

3.4 Optimal Control Problem Definition

The rendezvous optimal control formulation integrates safety regulation and energy-motion efficiency within the cost function, while enforcing ergonomic admissibility through terminal constraints.

The performance index proposed in this work is defined as

$$J^d(u^d(\cdot), t_f, \delta) = \int_{t_0}^{t_f} \left(\|u^d(t)\|_R^2 + \|v^d(t)\|_Q^2 + \|v^d(t) - v_{\text{safe}}(t)\dot{v}^d(t)\|_S^2 \right) dt. \quad (10)$$

where $R \succ 0$, $Q \succeq 0$, and $S \succeq 0$ are weighting matrices. The first term in the integral penalizes the control effort, the second promotes kinematic smoothness, and the third enforces compliance with the safe speed profile $v_{\text{safe}}(t)$. The unit vector $\hat{v}^d(t) = v^d(t)/\|v^d(t)\|$ represents the instantaneous heading direction of the drone. We remark that the quadratic structure of (10) facilitates indirect optimization and derivation of the necessary optimality conditions.

The feasible set of trajectories is determined by the translational dynamics introduced in (3), the safety regulation law (4), and the ergonomic terminal conditions given in (7)-(9). Control inputs are bounded within the admissible actuation set

$$u^d(t) \in \mathcal{U} \subset \mathbb{R}^3, \quad t \in [t_0, t_f], \quad (11)$$

where $\mathcal{U}$ denotes the set of admissible control functions $u^d : [t_0, t_f] \rightarrow \mathbb{R}^3$ accounting for thrust and attitude limitations of the aerial platform.

The complete human-drone rendezvous trajectory planning problem is then formulated as

$$\begin{aligned} \min_{u^d(\cdot) \in \mathcal{U}, t_f \geq t_0, \delta \in \mathcal{V}^*} \quad & J^d(u^d(\cdot), t_f, \delta) \\ \text{subject to} \quad & \\ \text{constraints (3), (7), (9), (11).} \end{aligned} \quad (12)$$

4. RENDEZVOUS PROBLEM REFORMULATION

Problem (12) is a nonconvex, free-final-time optimal control problem with control bounds and terminal conditions. The presence of the displacement vector δ introduces an implicit coupling between the control inputs and the drone's terminal state, preventing a direct solution via indirect methods. To address this issue, the rendezvous optimal control problem (12) is reformulated through a twofold approach. Section 4.1 introduces a parametric formulation that decouples the ergonomic constraint from the control variables by treating the displacement vector δ as a fixed parameter. The parametrization is adopted to enable the use of the classical Pontryagin theory (Kopp (1962)), which requires the terminal ergonomic constraint to be expressed as an equality condition for the analytical derivation of the necessary optimality conditions. Section 4.2 reformulates the resulting parametric optimal control problem in indirect form through the PMP, deriving the canonical state-costate equations and the corresponding BVP that characterize the optimal trajectories for each admissible $\delta \in \mathcal{V}^*$.

4.1 Parametric Optimization

To decouple the terminal ergonomic constraint from the optimization variables, the original problem (12) is reformulated by treating the ergonomic displacement vector δ as a fixed parameter. The resulting parametric reformulation is expressed as

$$\min_{\delta \in \mathcal{V}^*} \tilde{J}^d(\delta), \quad (13)$$

where the value function $\tilde{J}^d(\delta)$ is defined by the optimal control problem

$$\begin{aligned} \tilde{J}^d(\delta) = \min_{u^d(\cdot) \in \mathcal{U}, t_f \geq t_0} \quad & J^d(u^d(\cdot), t_f, \delta) \\ \text{subject to} \quad & \\ \text{constraints (3), (7), (9), (11).} \end{aligned} \quad (14)$$

The function $\tilde{J}^d(\delta)$ denotes the optimal value of the performance index in (10) (which aggregates the contributions penalizing control effort, promoting kinematic smoothness, and enforcing adherence to the safe speed profile) with δ treated as a fixed parameter.

4.2 Indirect Formulation and BVP Transformation

To determine $\tilde{J}^d(\delta)$ in (14), the problem is reformulated in indirect form by applying the PMP (Kopp (1962)). The resulting canonical system composed of the state, costate, and optimal control equations provides the necessary conditions for optimality corresponding to a fixed ergonomic displacement $\delta \in \mathcal{V}^*$.

These necessary conditions are derived by introducing the Hamiltonian associated with the cost function (10) and the corresponding costate dynamics. Let

$$\lambda^d(t) = (\lambda_p^d(t)^\top, \lambda_v^d(t)^\top)^\top \in \mathbb{R}^6 \quad (15)$$

denote the costate vector associated with the drone state $s^d(t)$, where $\lambda_p^d(t)$ and $\lambda_v^d(t)$ represent the costate variables associated with position and velocity, respectively.

The Hamiltonian is then defined as

$$\begin{aligned} H^d(s^d(t), u^d(t), \lambda^d(t), t) = & L^d(s^d(t), u^d(t), t) \\ & + (\lambda^d(t))^\top f^d(s^d(t), u^d(t), t), \end{aligned} \quad (16)$$

where the running cost is given by

$$\begin{aligned} L^d(s^d(t), u^d(t), t) = & \|u^d(t)\|_R^2 + \|v^d(t)\|_Q^2 \\ & + \|v^d(t) - v_{\text{safe}}(t)\hat{v}^d(t)\|_S^2. \end{aligned} \quad (17)$$

The stationarity condition for the PMP, i.e., $\frac{\partial H^d}{\partial u^d} = 0$, yields the optimal control law

$$u^{d,*}(t) = \text{proj}_{\mathcal{U}} \left(-\frac{1}{2m^d} R^{-1} \lambda_v^d(t) \right),$$

where $\text{proj}_{\mathcal{U}}(\cdot)$ denotes the projection onto the admissible actuation set $\mathcal{U}$, ensuring compliance with thrust limitations and dynamic feasibility.

Applying the canonical equations of the PMP, the indirect reformulation of the parametric problem (14) leads to the following two-point BVP:

$$\begin{cases} \dot{s}^d(t) = f^d(s^d(t), u^{d,*}(t), t), \\ \dot{\lambda}^d(t) = -\frac{\partial H^d}{\partial s^d}(s^d(t), u^{d,*}(t), \lambda^d(t), t) \end{cases} \quad (18a)$$

$$\begin{cases} p^d(t_0) = p^{d,0}, v^d(t_0) = v^{d,0}, \\ p^d(t_f) = p^h(t_f) + R^h(t_f)\delta, v^d(t_f) = 0, \\ H^d(t_f) = 0. \end{cases} \quad (18b)$$

Equations (18a) contain the state and costate dynamics, while eqs. (18b) specify the boundary conditions, including

the initial, terminal, and transversality conditions. Specifically, the transversality condition $H^d(t_f) = 0$ ensures consistency with the free-final-time formulation and concludes the canonical equation system.

Substituting the explicit expressions of $f^d(\cdot)$, $H^d(\cdot)$, and $u^{d,*}(\cdot)$ into (18a) results in the canonical state-costate dynamics:

$$\begin{cases} \dot{p}^d(t) = v^d(t), \\ \dot{v}^d(t) = -\frac{k^d}{m^d} v^d(t) + u^{d,*}(t), \\ \dot{\lambda}_p^d(t) = 2 \tilde{v}'_{\text{safe}}(d(t)) \hat{r}(t) \hat{v}^d(t)^\top S(v^d(t) - v_{\text{safe}}(t) \hat{v}^d(t)), \\ \dot{\lambda}_v^d(t) = \frac{k^d}{m^d} \lambda_v^d(t) - \lambda_p^d(t) - 2(Q + S) v^d(t) \\ \quad + 2S v_{\text{safe}}(t) \hat{v}^d(t) \end{cases} \quad (19)$$

where the unit relative position vector is defined as $\hat{r}(t) = \frac{p^d(t) - p^h(t)}{\|p^d(t) - p^h(t)\|}$, pointing from the human operator toward the drone. The derivative of the safe speed mapping $\tilde{v}'_{\text{safe}}(d(t))$ is expressed as a piecewise-constant function of the distance

$$\tilde{v}'_{\text{safe}}(d(t)) = \begin{cases} 0, & d(t) \leq d_{\text{stop}}, \\ \frac{v_{\text{cruise}}}{d_{\text{cruise}} - d_{\text{stop}}}, & d_{\text{stop}} < d(t) < d_{\text{cruise}}, \\ 0, & d(t) \geq d_{\text{cruise}}. \end{cases} \quad (20)$$

which is nonzero only in the intermediate range of the protective region. As for the state dynamics (i.e., the first and second equations in (19)) and costate dynamics (i.e., the second-last and last equations in (19)) they are obtained by differentiating the Hamiltonian (16) with state variables, respectively. The supplementary material, available at <https://github.com/SilviaProia95/Human-Drone-Rendezvous>, contains the full derivations.

Summing up, system (19) defines twelve first-order differential equations describing the optimal evolution of the drone's state and costate trajectories in continuous time, while (18b) provide the seven boundary conditions required to close the BVP. Note that the initial and terminal values of the costate $\lambda^d(t_f)$ are not prescribed and are determined implicitly by the numerical BVP solver (e.g., through the collocation methods, shooting or multiple-shooting procedures).

5. RENDEZVOUS REPLANNING APPROACH

In industrial warehouses, human motion usually follows fixed and structured paths, such as aisles or corridors connecting storage zones and palletizing areas. The corresponding trajectories can be predefined and known in advance, allowing the drone to plan the rendezvous within a spatially constrained environment. Nevertheless, the operator's walking speed may vary according to task execution, carried load, and environmental conditions, resulting in a nonuniform progression of the human position along the predefined path. The corresponding motion evolves with uncertain timing, which induces time-varying boundary conditions in (18b).

Algorithm 1 Free-Final-Time Shrinking-Horizon Rendezvous Replanning

Input: t_0 , $s^{d,0}$, and $p^h(t), \psi^h(t), t \geq t_0$

Parameters: $\varepsilon, \Delta t$

Output: $u^{d,*}(t)$

```

1:  $t_j = t_0$ 
2: while  $\|p^d(t_j) - p^h(t_j)\| > \varepsilon$  do
3:    $t_0 = t_j$ 
4:   Evaluate  $s^d(t_j)$ 
5:   Update  $p^h(t), \psi^h(t), t \geq t_0$ 
6:   Solve (13) using (18b) – (19)
7:   Apply  $u^{d,*}(t), t \in [t_j, t_j + \Delta t]$ 
8:    $t_j \leftarrow t_j + \Delta t$ 
9: end while
```

To preserve safety, ergonomics compliance, and energy efficiency as the human pose evolves, the rendezvous optimal control problem (13)-(18b)-(19) is iteratively solved within a receding-horizon scheme. At each replanning cycle, the problem is integrated from the current time instant t_j over the optimal prediction horizon T_j , which is determined as part of the solution. As the drone approaches the human operator, the prediction horizon T_j is progressively shortened, leading to a shrinking-horizon behavior.

Algorithm 1 – *Free-Final-Time Shrinking-Horizon Rendezvous Replanning* iteratively updates the optimal drone trajectory in response to the evolving human motion. The algorithm takes as input the drone initial state $s^d(t_0)$, together with the human position $p^h(t)$ and heading $\psi^h(t)$, with $t \geq t_0$. At each replanning cycle, the current drone state $s^d(t_j)$ and human motion variables $p^h(t), \psi^h(t), t \geq t_0$ are evaluated and used to update the terminal boundary conditions of the BVP (18b)(Algorithm 1, lines 4-5). The rendezvous optimal control problem (13)-(18b)-(19) is then solved (Algorithm 1, line 6), providing the optimal control input $u_j^{d,*}(t)$. The control is applied over the interval $[t_j, t_j + \Delta t]$ (Algorithm 1, line 7), after which the next replanning instant is updated as $t_j = t_j + \Delta t$ (Algorithm 1, line 8). The process is repeated until the rendezvous condition $\|p^d(t_j) - p^h(t_j)\| \leq \varepsilon$ is satisfied, where ε denotes the distance tolerance defining the terminal proximity between the drone and the human.

6. NUMERICAL EXPERIMENTS

6.1 Setup

The simulations aim at validating the performance of the proposed human-drone rendezvous control approach in a warehouse environment. The considered scenario reproduces a collaborative pick-and-delivery operation, where drones assist human operators during palletizing and material-handling tasks. All simulations are executed in MATLAB R2024b, where the control framework integrates the nonlinear optimizer `fmincon` for the parametric optimization with the collocation-based solver `bvp4c` for the numerical solution of the free-final-time BVP.

The quadrotor operates within the ENU frame centered at the lower-left vertex of the warehouse workspace. It is modeled as a translational point mass with viscous damping coefficient $k^d = 0.05$ and total mass $m^d = 1.00$ kg. The

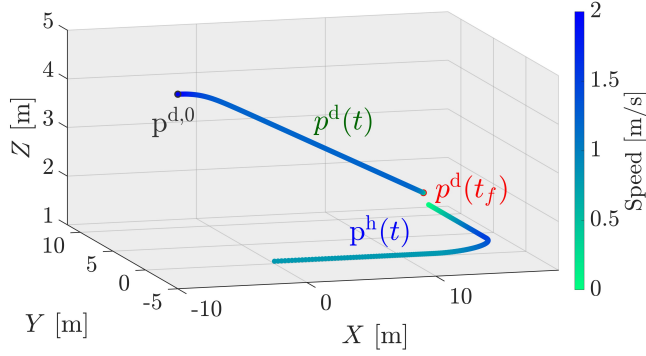


Fig. 2. Drone and human trajectories during the rendezvous maneuver with colormap based on speed.

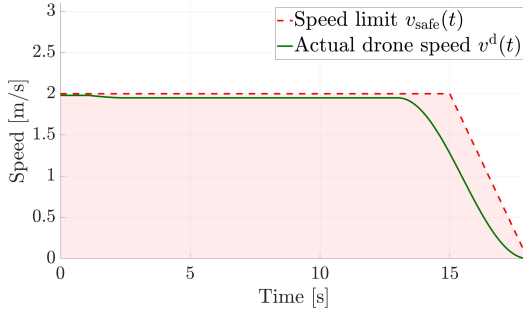


Fig. 3. Comparison between the desired safe speed profile and the actual drone speed.

initial position is set to $p^{d,0} = (-10.00, -5.00, 5.00)^\top$ m and the initial velocity to $v^{d,0} = (2.00, 0.00, 0.00)^\top$ m/s, corresponding to a forward motion along the x direction. The admissible control set is defined as $\mathcal{U} = \{u^d \in \mathbb{R}^3 \mid \|u^d\|_\infty \leq 3 \text{ m/s}^2\}$, which reflects the drone's actuation limits.

The human operator is a right-handed adult male at the 50th percentile of height and follows a time-parametrized trajectory $p^h(t)$ composed of an initial rectilinear segment along the X -axis, a smooth circular transition, and a subsequent translation along the Y -axis, representative of a collaborative pick-and-delivery task in an industrial warehouse. The walking speed $v^h(t)$ varies between 0.8 m/s and 1.5 m/s, consistently with typical ranges observed in human locomotion during collaborative operations.

The ergonomic volume $\mathcal{V}^*$ is defined in the human reference frame $\{F^h(t)\}$ and bounded within the ranges $[-0.20, 0.10]$ m, $[0.35, 0.50]$ m, and $[1.25, 1.35]$ m along the forward, lateral, and vertical directions, as determined through the RULA ergonomic analysis in Proia et al. (2023, 2025). Safety constraints are implemented by means of a distance-dependent speed limit $v_{\text{safe}}(t)$, characterized by $d_{\text{stop}} = 0.10$ m, $d_{\text{cruise}} = 3.00$ m, and $v_{\text{cruise}} = 2.00$ m/s.

6.2 Results

The following results illustrate the overall behavior of the proposed control framework, emphasizing the system's capability to adapt to the human motion while preserving safety, ergonomics, and efficiency compliance.

The rendezvous operation performed by the quadrotor, along with the trajectory generated by the BVP-based

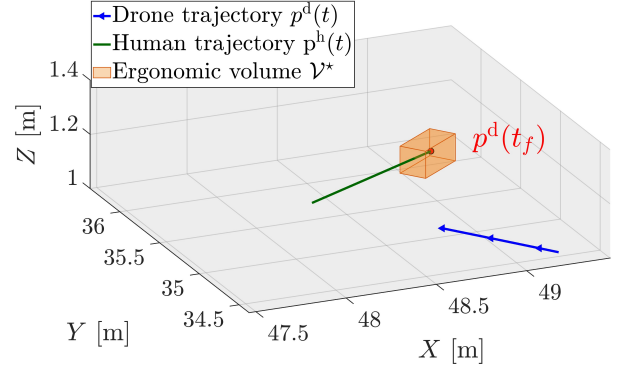


Fig. 4. Zoomed view of the final rendezvous showing the drone reaching the ergonomic volume.

Table 1. Performance comparison between the reference method and the proposed solution.

Metric	Unit	Proia et al. (2025)	Proposed Solution	Improvement [%]
Mission Time	[s]	6.12	5.28	↓13.7
Drone Path Length	[m]	11.86	9.64	↓18.7
Mean Control Effort	[-]	1.57	0.16	↓89.8
Kinetic Energy	[-]	15.21	8.98	↓40.9
Trajectory Smoothness Index	[-]	11.27	8.04	↓28.6
Final Position Error	[m]	1.38	0.44	↓68.1

controller, is shown in Fig. 2. As highlighted by the colormap, the human operator exhibits a varying walking speed along the path, slowing down during the change of direction and accelerating once the operator is aligned with the final path. The controller continuously replans, at each $\Delta t = 1$ s, to account for the nonuniform human motion, ensuring a smooth trajectory and consistent convergence despite the time-varying boundary conditions.

The effectiveness of the proposed strategy to ensure safety is highlighted in Fig. 3, which compares the imposed safe speed profile $v_{\text{safe}}(t)$ with the actual drone speed obtained from the rendezvous problem solution. The drone speed consistently remains below the admissible safety limit, confirming full compliance with the distance-dependent regulation law and demonstrating that the controller inherently enforces safe behavior during the entire rendezvous maneuver. In addition to safety, the proposed formulation enforces ergonomics compliance as part of the terminal boundary constraint. Figure 4 shows a zoomed view of the final rendezvous trajectories of the drone and the operator, together with the ergonomic interaction volume at the final time. The terminal position lies within $\mathcal{V}^*$, corresponding to the comfortable reach region identified through the RULA-based assessment. The optimal HDI point, obtained from the rendezvous control problem solved via the MATLAB function `fmincon-bvp`, results in $p^d(t_f) = (-0.07, 0.43, 1.31)^\top$ m. This outcome confirms that the quadrotor converges to an ergonomically admissible and energy-efficient configuration, ensuring both human comfort and dynamic feasibility.

To assess the performance of the proposed control strategy, a comparative analysis is performed with respect to the descent control method presented in Proia et al. (2025), which, to the best of our knowledge, represents one of the most relevant references in the literature on HDI in shared workspaces. In Proia et al. (2025), trajectory

generation is based on Hermite interpolation, and motion tracking is managed through a receding-horizon LQR (rhLQR) controller. Safety constraints are enforced only during the planning phase, whereas ergonomic feasibility is represented by a fixed terminal configuration. In contrast, the BVP-based formulation enables simultaneous handling of safety, ergonomics, and energy objectives by embedding all constraints within a unified optimal control problem. The quantitative comparison is based on the following performance metrics:

- *Mission Time*: total duration required to complete the rendezvous maneuver;
- *Drone Path Length*: total distance traveled by the quadrotor along the rendezvous trajectory;
- *Mean Control Effort*: average actuator workload over the mission duration;
- *Kinetic Energy*: cumulative mechanical energy associated with the drone's translational motion;
- *Trajectory Smoothness Index*: measure of control signal continuity, based on the time derivative of the control inputs;
- *Final Position Error*: Euclidean distance between the terminal drone position and the target point.

The numerical results are reported in Table 1, where the proposed BVP-based framework is evaluated against the reference Hermite-rhLQR method from Proia et al. (2025). The comparison highlights a marked improvement across all performance indicators. In particular, the mean control effort exhibits a significant decrease (around 90%), indicating a lower actuation demand and a smoother dynamic behavior. The kinetic energy and trajectory smoothness index are also reduced, confirming the generation of dynamically consistent and energy-efficient motions. Both mission time and drone path length decrease, showing that jointly addressing safety and energy objectives leads to smoother and more efficient task execution. Accurate terminal convergence to the ergonomic configuration is achieved, as evidenced by the negligible final position error. The reduction in control and energetic indicators demonstrates that the proposed framework produces smoother, energy-efficient, and human-compliant trajectories suitable for HDI.

7. CONCLUSION

A novel receding-horizon Boundary Value Problem (BVP) formulation has been presented for autonomous human-drone rendezvous in industrial Human-Drone Interaction (HDI). The method unifies safety, ergonomic, and energy-efficiency objectives within an indirect optimal control framework, ensuring terminal constraint satisfaction and real-time adaptability through a shrinking-horizon strategy. Simulation results in a warehouse scenario confirm that the controller ensures safe and ergonomic co-operation, producing smooth and energy-efficient trajectories with accurate terminal convergence. Future works will account for predictive modeling of operator behavior, adaptive triggering of the replanning phase, and enhanced robustness in more complex rendezvous scenarios.

REFERENCES

Avvakumov, S. and Kiselev, Y.N. (2000). Boundary value problem for ordinary differential equations with applications to optimal control. *Spectral and evolution problems*, 10, 147–155.

- Das, A., Subbarao, K., and Lewis, F. (2009). Dynamic inversion with zero-dynamics stabilisation for quadrotor control. *IET control theory & applications*, 3(3), 303–314.
- Kopp, R.E. (1962). Pontryagin maximum principle. In *Mathematics in Science and Engineering*, volume 5, 255–279. Elsevier.
- Li, Q., Yuan, J., Zhang, B., and Gao, C. (2017). Model predictive control for autonomous rendezvous and docking with a tumbling target. *Aerospace Science and Technology*, 69, 700–711.
- Lucci, N., Lacevic, B., Zanchettin, A.M., and Rocco, P. (2020). Combining speed and separation monitoring with power and force limiting for safe collaborative robotics applications. *IEEE Robotics and Automation Letters*, 5(4), 6121–6128.
- Luo, Y., Zhang, J., and Tang, G. (2014). Survey of orbital dynamics and control of space rendezvous. *Chinese Journal of Aeronautics*, 27(1), 1–11.
- McAtamney, L. and Corlett, E.N. (1993). Rula: a survey method for the investigation of work-related upper limb disorders. *Applied Ergonomics*, 24(2), 91–99.
- Miele, A., Weeks, M., and Ciarcia, M. (2007). Optimal trajectories for spacecraft rendezvous. *Journal of optimization theory and applications*, 132(3), 353–376.
- Mirri, S., Prandi, C., and Salomoni, P. (2019). Human-drone interaction: state of the art, open issues and challenges. In *Proceedings of the ACM SIGCOMM 2019 Workshop on Mobile AirGround Edge Computing, Systems, Networks, and Applications*, 43–48.
- Proia, S., Carli, R., Cavone, G., and Dotoli, M. (2021). Control techniques for safe, ergonomic, and efficient human-robot collaboration in the digital industry: A survey. *IEEE Transactions on Automation Science and Engineering*, 19(3), 1798–1819.
- Proia, S., Cavone, G., Scarabaggio, P., Carli, R., and Dotoli, M. (2023). Safety compliant, ergonomic and time-optimal trajectory planning for collaborative robotics. *IEEE Transactions on Automation Science and Engineering*.
- Proia, S., Cavone, G., Scarabaggio, P., Carli, R., and Dotoli, M. (2025). An integrated control framework for safe and ergonomic human-drone interaction in industrial warehouses. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.
- Quartullo, R., Bianchini, G., Garulli, A., and Giannitrapani, A. (2024). Robust time-optimal model predictive control for rendezvous with a moving target. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, 7510–7515. IEEE.
- Sabatino, F. (2015). Quadrotor control: modeling, nonlinear control design, and simulation.
- Wang, X. (2009). Solving optimal control problems with matlab: Indirect methods. *ISE Dept., NCSU, Raleigh, NC*, 27695.
- Wawrla, L., Maghazei, O., and Netland, T. (2019). Applications of drones in warehouse operations. *Whitepaper. ETH Zurich, D-MTEC*.
- Wei, Z., Yang, J., Wen, H., Jin, D., and Chen, T. (2025). Nonlinear optimal control for spacecraft rendezvous and docking using symplectic numerical method. In *Actuators*, volume 14, 75. MDPI AG.